



C23-M-403

23447

BOARD DIPLOMA EXAMINATION, (C-23)

MARCH/APRIL—2025

DME - FOURTH SEMESTER EXAMINATION

THERMAL ENGINEERING - II

Time : 3 Hours]

[Total Marks : 80

PART—A

3×10=30

Instructions : (1) Answer **all** questions.

(2) Each question carries **three** marks.

(3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

1. Mention any three differences between twostroke and fourstroke engines.
2. Write any three functions of lubrication in IC engines.
3. What is the purpose of engine testing?
4. Define (a) Indicated power and (b) Mechanical efficiency.
5. Write the classification of reciprocating air compressor based on any three criteria.
6. State any three advantages of multistage compression over singlestage compression.
7. Write the classification of Gas turbines.
8. List various fuels used in jet engines.
9. Write the functions of (a) Clutch and (b) gear box

10. Mention any three features of BS-VI regulations.

PART—B

5×10=50

Instructions : (1) Answer **five** questions.

(2) Each question carries **ten** marks.

(3) Answers should be comprehensive and criterion for valuation is the content but not the length of the answer.

11. Explain the working of fourstroke petrol engine with a neat sketch.

12. Explain the working of Battery ignition system in the SI engines with a line diagram.

13. A four-cylinder, 4-stroke petrol engine having 70 mm bore and 90 mm stroke is tested at full throttle at constant speed. The fuel supply is fixed at 0.065 kg/min and plugs of the 4 cylinders are successively short circuited without change of speed. Brake torque being correspondingly adjusted. The calorific value of fuel is 42000 kJ/kg. Following power requirements are made under different conditions :

BP with all cylinders firing = 11.9kW

BP with cylinder No.1 short circuited = 8.46 kW

BP with cylinder No.2 short circuited = 8.57 kW

BP with cylinder No.3 short circuited = 8.61 kW

BP with cylinder No.4 short circuited = 8.5 kW

Calculate : (a) Indicated power, (b) Friction power, (c) Mechanical efficiency, (d) Brake thermal efficiency and (e) Indicated thermal efficiency.

14. A two-stage air compressor is used to compress 1 kg of free air from 1 bar and 28°C to 30 bar. The value of $n = 1.25$ and $R = 0.29$ kJ/kg-K. Determine the intermediate pressure, (b) work required for the best performance, (c) work for a corresponding single-stage compressor, and (d) percentage saving in work in two-stage compressor.

- 15.** Explain the working of Centrifugal compressor with the help of neat sketch.
- 16.** Explain the working of Open cycle gas turbine power plant the help of neat sketch.
- 17.** (a) Explain the of Turbo Jet engine with a line diagram.
(b) Explain the working of Exhaust Gas Re-circulation (EGR) with the help of line diagram.
- 18.** Explain the working of Differential with the help of neat sketch.

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